

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension I

Year 12 Higher School Certificate
Trial Examination Term 3 2013

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided separately
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination

Total marks – 70

Section I Pages 3 – 6

10 marks

Attempt Questions 1 – 10

Answer on the Objective Response Answer Sheet provided

Section II Pages 7 – 11

60 marks

Attempt Questions 11 – 14.

Start each question in a new writing booklet.

Write your student number on every writing booklet.

Question	1-10	11	12	13	14	Total
Total	/10	/15	/15	/15	/15	/70

This assessment task constitutes 45% of the Higher School Certificate Course School Assessment

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Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 10

- 1 A polynomial equation has roots α , β and γ , where:
 $\alpha + \beta + \gamma = -3$, $\alpha\beta + \alpha\gamma + \beta\gamma = -2$ and $\alpha\beta\gamma = 4$.
Which polynomial equation has the roots α , β and γ ?

- (A) $x^3 - 3x^2 - 2x + 4 = 0$
(B) $x^3 + 3x^2 - 2x - 4 = 0$
(C) $x^3 + 3x^2 + 2x + 4 = 0$
(D) $x^3 - 3x^2 + 2x - 4 = 0$

- 2 The solution to $\frac{4}{x-3} \leq 2$ is:

- (A) $3 \leq x \leq 5$
(B) $3 < x \leq 5$
(C) $x < 3$ or $x \geq 5$
(D) $x \leq 3$ or $x \geq 5$

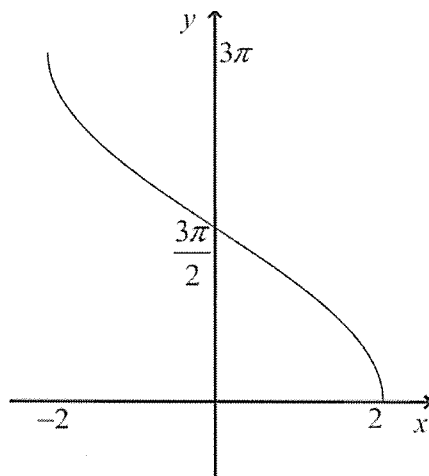
- 3 $\int \cos^2 4x \, dx =$

- (A) $\frac{1}{2}x + \frac{1}{16}\sin 8x + C$
(B) $\frac{1}{2}x - \frac{1}{16}\sin 8x + C$
(C) $\frac{1}{2}x + \frac{1}{8}\sin 8x + C$
(D) $\frac{1}{2}x - \frac{1}{8}\sin 8x + C$

4 If $P(x) = (x+2)(x+k)$ and if the remainder when $P(x)$ is divided by $(x-1)$ is 12, then:

- (A) $k = 2$
- (B) $k = 3$
- (C) $k = 6$
- (D) $k = 11$

5 Which function best describes the graph below?

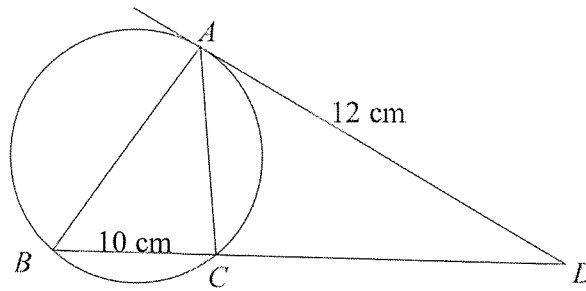


- (A) $y = 2 \cos^{-1} 3x$
- (B) $y = 2 \cos^{-1} \frac{3x}{2}$
- (C) $y = 3 \cos^{-1} 2x$
- (D) $y = 3 \cos^{-1} \frac{x}{2}$

6 If the function f is defined by $f(x) = x^5 - 1$, then the inverse function of f , is defined by $f^{-1}(x) =$

- (A) $\sqrt[5]{x} - 1$
- (B) $\sqrt[5]{x-1}$
- (C) $\sqrt[5]{x} + 1$
- (D) $\sqrt[5]{x+1}$

- 7 ABC is a triangle inscribed in a circle. The tangent to the circle at A meets BC produced at D where $BC = 10$ and $AD = 12$. What is the length of CD ?



NOT TO
SCALE

- (A) 6 cm
(B) 7 cm
(C) 8 cm
(D) 9 cm

8 $\int \frac{x^2}{e^{x^3}} dx =$

- (A) $-\frac{1}{3e^{x^3}} + C$
(B) $-\frac{1}{3}e^{x^3} + C$
(C) $-\frac{1}{3}\ln e^{x^3} + C$
(D) $\frac{1}{3}\ln e^{x^3} + C$

- 9 Consider the curve defined by the parametric equations $x = \frac{1}{t}$ and $y = \frac{t}{t+1}$.

The graph of $y = f(x)$ would have asymptotes:

- (A) $x = 0$ only
(B) $x = 1, y = -1$
(C) $x = -1$ only
(D) $x = -1, y = 0$

- 10** The velocity, v metres per second, of a particle moving in simple harmonic motion along the x axis is given by the equation $v^2 = 36 - 4x^2$.

What is the amplitude, in metres of the motion of the particle?

- (A) 3
- (B) 2
- (C) 6
- (D) 4

End of Section I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations

Question 11 (15 marks) Start a new writing booklet

(a) Evaluate $\int_0^2 \frac{dx}{\sqrt{16-x^2}}$. 2

(b) Differentiate $3x^2 \ln x$, for $x > 0$. 2

(c) Find the acute angle between the lines $x + 2y - 5 = 0$ and $y = 4x + 5$, giving your answer 3
correct to the nearest minute.

(d) Use the substitution $u = e^x$ to find $\int \frac{e^x}{1+e^{2x}} dx$. 3

(e) The staff in a school office consists of 5 males and 8 females. 2
How many committees of 5 staff can be chosen that contain exactly 3 females?

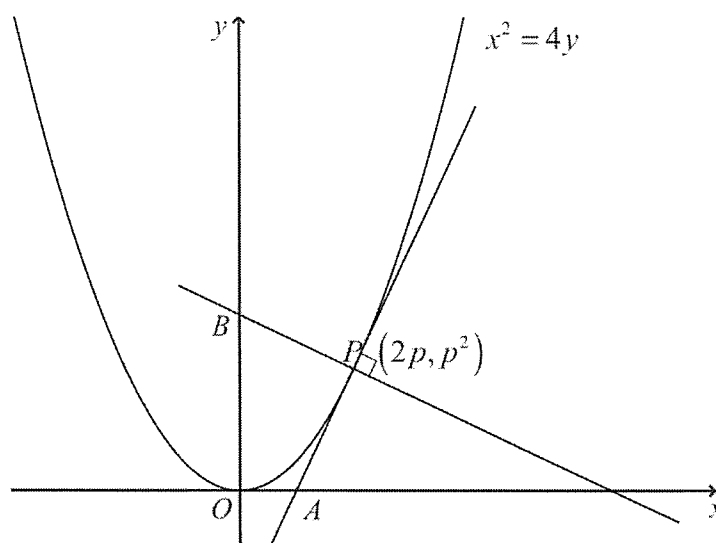
(f) Use the binomial theorem to find the term independent of x in the expansion of 3
 $\left(2x - \frac{1}{x^2}\right)^{12}$.

Question 12 (15 marks) Start a new writing booklet

- (a) Use mathematical induction to prove that $n! > 2^n$ for integer $n \geq 4$.

3

- (b) The diagram below shows the graph of the parabola $x^2 = 4y$.
The tangent cuts the parabola at $P(2p, p^2)$, $p > 0$, cuts the x axis at A .
The normal to the parabola at P cuts the y axis at B .



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- (i) Show that the equation of the tangent AP is $y = px - p^2$.

2

- (ii) Show that the equation of the normal PB is $x + py = p^3 + 2p$.

1

- (iii) Find the coordinates of A and B .

2

- (iv) Let C divide the interval AB in the ratio 2:1.

3

Find the Cartesian equation of the locus of C , giving any domain restrictions.

- (c) Consider the function $f(x) = 1 + \cos^{-1}(2x-1) - 2\cos^{-1}\sqrt{x}$ for $0 \leq x \leq 1$.

- (i) Show that $f'(x) = 0$ for $0 \leq x \leq 1$.

3

- (ii) Sketch the graph of $y = f(x)$ for $0 \leq x \leq 1$.

1

Question 13 (15 marks) Start a new writing booklet

- (a) A particle is moving in simple harmonic motion has its acceleration given by

$$\frac{d^2x}{dt^2} = -25x, \text{ where } x \text{ metres is the displacement of the particle after } t \text{ seconds.}$$

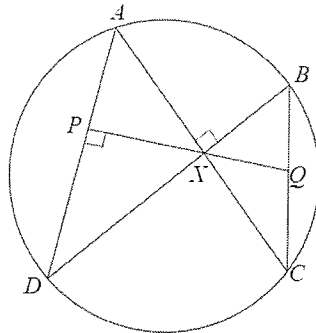
Initially, the particle's acceleration is 50 ms^{-2} and after $\frac{\pi}{6}$ seconds, the particle's velocity is -10 ms^{-1} .

- (i) Find the period of the motion. 1
- (ii) Show that $x = a \sin(5t - \alpha)$ is a possible equation of motion for this particle, 2
where a and α are positive constants and α is acute.
- (iii) Show that the amplitude of the motion is 4 metres. 2
- (iv) Find the value of α . 1
- (v) Find the greatest speed of the particle and where the particle reaches this speed. 2
- (vi) How many times does the particle change direction in the first 2 seconds? 2
- (b) Let $(2 + 3x)^7 = \sum_{k=0}^7 t_k x^k$
- (i) Write down an expression for t_k . 1
- (ii) Hence show that $\frac{t_{k+1}}{t_k} = \frac{21-3k}{2k+2}$ where $0 < k < 7$. 2
- (iii) Hence, or otherwise, find the greatest coefficient in the expansion of $(2 + 3x)^7$. 2

Question 14 (15 marks) Start a new writing booklet

- (a) The diagram below shows points A, B, C and D on a circle. The lines AC and BD are perpendicular and meet at X .

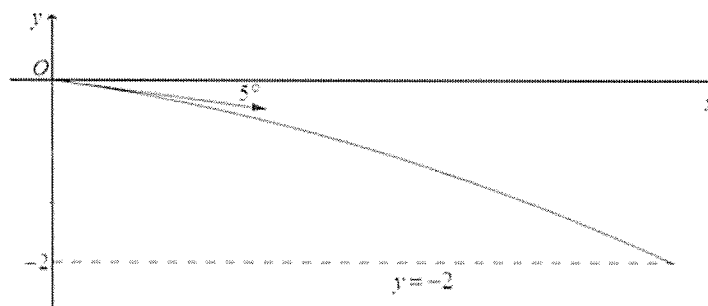
The perpendicular to AD through X meets AD at P and BC at Q .



NOT TO
SCALE

Copy or trace this diagram into your writing booklet.

- (i) Prove that $\angle QXB = \angle QBX$. 3
- (ii) Prove that PQ bisects BC . 2
- (b) A cricket ball leaves a bowler's hand 2 metres above the ground with a velocity of 30 ms^{-1} at an angle of projection of 5° **below** the horizontal, as shown below.



NOT TO
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Using the origin as the point where the ball leaves the bowlers hand, the coordinates of the ball at time t are given by:

$$x = 30t \cos 5^\circ$$

$$y = -30t \sin 5^\circ - 5t^2$$

(Do not prove these results)

- (i) Find the time it takes for the ball to strike the ground. 2
- (ii) Calculate the angle at which the ball strikes the ground. 2
- (iii) Show the motion of the ball is parabolic, even though it is projected at an angle 2
- below the horizontal.

Question 14 continues on page 11

Question 14 (continued)

- (c) A television satellite tower stands on a large area of flat ground.

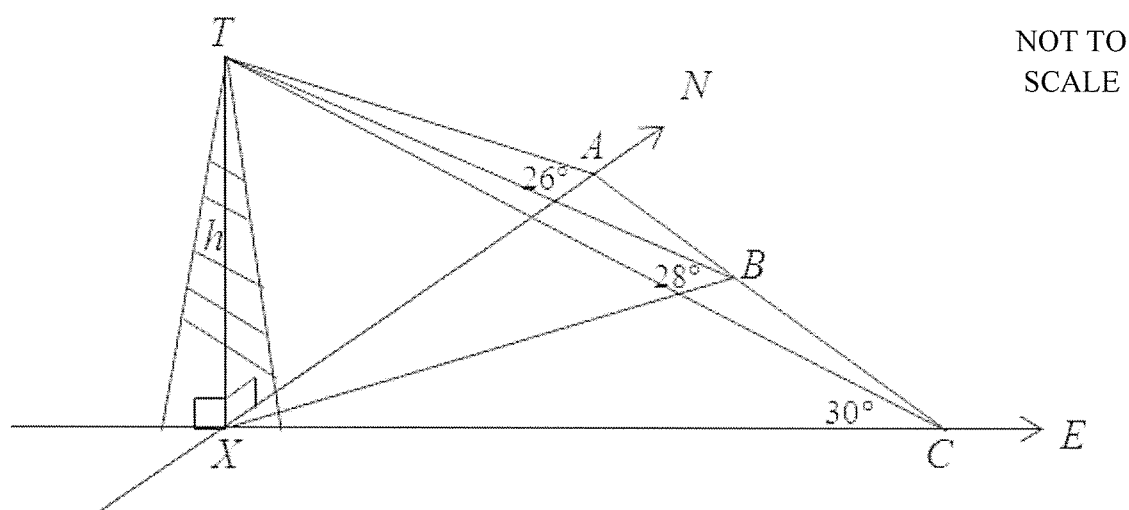
Three maintenance workers, A, B and C, are observing the tower.

Worker A is due north of the tower.

Worker C is due East of the tower.

Worker B is on the line of sight-from A to C (A, B and C are collinear).

The angles of elevation of the top of the tower from A, B and C are 26° , 28° and 30° respectively.



- | | |
|--|----------|
| (i) Find $\angle XAC$, correct to one decimal place. | 1 |
| (ii) Find $\angle ABX$, correct to the nearest degree. | 2 |
| (iii) Hence, find the bearing of Worker B from the base of the tower X, correct to the nearest degree. | 1 |

End of Paper

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Ext 1 Trial Solutions 2013 HGHS

Multiple Choice

1. $-\frac{b}{a} = -3$ $\frac{c}{a} = -2$ $-\frac{d}{a} = 4$
 $\therefore a=1$ $b=3$ $c=-2$ $d=-4$ (B)

2. $\frac{4}{x-3} \leq 2$

$4(x-3) \leq 2(x-3)^2$

$4(x-3) - 2(x-3)^2 \leq 0$

$(x-3)[4 - 2(x-3)] \leq 0$

$(x-3)(10-2x) \leq 0$

$2(x-3)(5-x)$

$x < 3, x \geq 5$ (C)

3. $\int \cos^2 4x \, dx = \frac{1}{2} \int [\cos 8x + 1] \, dx$

$= \frac{1}{2} \left[\frac{\sin 8x}{8} + x \right]$

$= \frac{x}{2} + \frac{\sin 8x}{16} + c$

(A)

4. $p(1) = 12$

$\therefore (1+2)(1+k) = 12$

$3+3k = 12$

$3k = 9$

$k = 3$

(B)

5. $-1 \leq \frac{x}{2} \leq 1$

$-2 \leq x \leq 2$

$\therefore$ (D) is only option

6. $f(x) = x^5 - 1$

$\therefore$ for inverse

$x = f(x)^5 - 1$

$\therefore f(x)^5 = x+1$

$\therefore f(x) = \sqrt[5]{x+1}$

(D)

7. $AD^2 = BD \times CD$

$12^2 = (10+x)x$

$144 = 10x + x^2$

$0 = x^2 + 10x - 144$

$0 = (x+18)(x-8)$

$\therefore x = -18, 8$

$x = 8$ as $x > 0$

8. $\int \frac{x^2}{e^{x^3}} \, dx = \int x^2 e^{-x^3} \, dx$

$= -\frac{1}{3} e^{-x^3} + c$

$= -\frac{1}{3} e^{-x^3} + c$ (A)

9. $x = \frac{1}{t}$ then $y = \frac{1}{x} = t$

$= \frac{1}{1+x}$

$\therefore x \neq -1$ and $y = 0$ (D)

10. $v^2 = 36 - 4x^2$

$= 4(9 - x^2)$

$= n^2(a^2 - x^2)$

(A)

where $n=2$ and $a=3$

(1)

Question 11

(a) $\int_0^2 \frac{dx}{\sqrt{16-x^2}} = \left[\sin^{-1} \frac{x}{4} \right]_0^2$
 $= \sin^{-1} \frac{1}{2} - \sin 0$
 $= \frac{\pi}{6} - 0$
 $= \frac{\pi}{6}$

(b) $\frac{d}{dx} (3x^2 \ln x) = 3x^2 \cdot \frac{1}{x} + 6x \cdot \ln x$
 $= 3x + 6x \ln x$
 OR
 $= 3x(1 + 2 \ln x)$

(c) $x+2y-5=0$ and $4x-y+5=0$
 $m_1 = -\frac{1}{2}$ $m_2 = 4$

$\therefore \tan \alpha = \left| \frac{4 - (-\frac{1}{2})}{1 + 4 \cdot (-\frac{1}{2})} \right|$

$= \left| \frac{\frac{9}{2}}{-1} \right|$

$= 4\frac{1}{2}$

$\alpha = 77^\circ 28'$

(d) $\int \frac{e^x}{1+e^{2x}} \, dx = \int \frac{u}{1+u^2} \cdot \frac{du}{u}$ $u = e^x$
 $\therefore \frac{du}{dx} = e^x$
 $= \tan^{-1} u + c$
 $= \tan^{-1} e^x + c$
 $dx = \frac{du}{e^x}$
 $= \frac{du}{u}$

(2)

$$(e.) {}^8C_3 \times {}^5C_2 = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} \times \frac{5 \times 4}{2 \times 1}$$

$$= 560$$

$\therefore$ 560 different committees of 5 containing exactly 3 females could be chosen.

$$(f) \left(2x - \frac{1}{x^2}\right)^{12} \text{ has general term } {}^{12}C_r (2x)^{12-r} \left(-\frac{1}{x^2}\right)^r$$

$$\therefore \text{power of } x \text{ function} = 12 - r - 2r$$

$$= 12 - 3r$$

$\therefore$ constant term has power of zero

$$\therefore 12 - 3r = 0$$

$$\therefore r = 4$$

$$\text{If } r = 4 \quad {}^{12}C_4 (2x)^{12-4} \left(-\frac{1}{x^2}\right)^4 = 495 \times 2^8 \times x^8 \times -\frac{1}{x^8}$$

$$= -126720$$

Question 12

(a) RTP $n! > 2^n$ for $n \geq 4$

Step 1 Prove for $n = 4$

$$\text{LHS} = 4! \quad \text{RHS} = 2^4$$

$$= 24 \quad = 16$$

$\therefore$ true for $n = 4$

Step 2 If result is true for $n = k$
then $k! > 2^k$

Step 3 RTP for $n = k+1$ i.e. $(k+1)! > 2^{k+1}$

$$\text{LHS} = (k+1)!$$

$$= (k+1)k!$$

$$> (k+1)2^k$$

$$= k \cdot 2^k + 2^k$$

③

for $k \geq 4$

$$> 2^k + 2^k$$

$$= 2 \cdot 2^k$$

$$\therefore (k+1)! > 2^{k+1} \text{ for } k \geq 4$$

Step 4 Since the result is true for $n = 4$, it is therefore true for $n = 5$. Hence since it is true for $n = 5$ it is true for $n = 6$ and so on.

Therefore by the process of Mathematical Induction it is true for all integer k values greater than 4

$$(b) x^2 = 4y \quad x = 2p, y = p^2$$

$$(i) \frac{dx}{dp} = 2 \quad \frac{dy}{dp} = 2p \quad \therefore \frac{dy}{dx} = \frac{2p}{2} = p$$

$$\therefore \text{Tangent } y - p^2 = p(x - 2p)$$

$$y - p^2 = px - 2p^2$$

$$y = px - p^2 \text{ as required}$$

$$(ii) \text{ Normal } y - p^2 = -\frac{1}{p}(x - 2p)$$

$$py - p^3 = -x + 2p$$

$$x + py = p^3 + 2p \text{ as required}$$

$$(iii) A(y=0) \text{ on tangent } 0 = px - p^2$$

$$px = p^2$$

$$x = p$$

$$\therefore A(p, 0)$$

$$B(x=0) \text{ on normal } py = p^3 + 2p$$

$$y = p^2 + 2$$

$$\therefore B(0, p^2 + 2)$$

④

(iv) $A(p, 0)$, $B(0, p^2+2)$

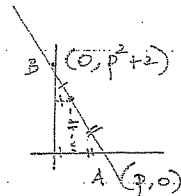
$\therefore C\left(\frac{p}{3}, \frac{2p^2+4}{3}\right)$ By the geometry

$\therefore x = \frac{p}{3}$ into $y = \frac{2(3x)^2 + 4}{3}$

$= \frac{18x^2 + 4}{3}$

$y = 6x^2 + \frac{4}{3}$

$\therefore C$ is a parabola with Domain $x > 0$ (since $p > 0$)
and range $y \geq \frac{4}{3}$, Vertex $(0, \frac{4}{3})$



(c) (i) $f(x) = \cos^{-1}(2x-1) - 2\cos^{-1}\sqrt{x} + 1$ $0 \leq x \leq 1$

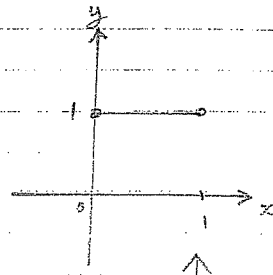
$f'(x) = \frac{-1}{\sqrt{1-(2x-1)^2}} \times 2 - \frac{2x-1}{\sqrt{1-(\sqrt{x})^2}} \times \frac{1}{2\sqrt{x}}$

$= \frac{-2}{\sqrt{1-(4x^2-4x+1)}} + \frac{1}{\sqrt{1-x^2}} \times \frac{1}{\sqrt{x}}$

$= \frac{-2}{\sqrt{4x-4x^2}} + \frac{1}{\sqrt{x}\sqrt{1-x^2}}$

$= \frac{-2}{2\sqrt{x}\sqrt{1-x^2}} + \frac{1}{\sqrt{x}\sqrt{1-x^2}}$

$= 0$ as required



(ii) $f(x)$ is horizontal since $f'(x) = 0$ for $0 \leq x \leq 1$

$f(0) = \cos^{-1}(-1) - 2\cos^{-1}(0) + 1$
 $= \pi - 2 \times \frac{\pi}{2} + 1 = 1$ (5)

Question 13

(a) $\ddot{x} = -25x$

(i) period $= \frac{2\pi}{\omega} = \frac{2\pi}{5}$ seconds

(ii) $x = a \sin(5t - \alpha)$

$\dot{x} = 5a \cos(5t - \alpha)$

$\ddot{x} = -25a \sin(5t - \alpha)$

$= -5^2(\sin(5t - \alpha))$

$= -5^2 x$

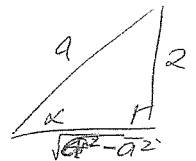
(i) $\ddot{x} = -25a \sin(-\alpha) = 50$

$\sin(-\alpha) = \frac{-2}{a}$

$\sin \alpha = \frac{2}{a}$ (1)

$\ddot{x} = 5a \cos(5\pi/6 - \alpha) = -10$

$a \cos(5\pi/6 - \alpha) = -2$ (2)



$a[\cos 5\pi/6 \cos \alpha + \sin 5\pi/6 \sin \alpha] = -2$

$a\left[-\frac{\sqrt{3}}{2} \times \frac{\sqrt{a^2-4}}{a} + \frac{1}{2} \times \frac{2}{a}\right] = -2$

$-\frac{\sqrt{3}\sqrt{a^2-4}}{2a} + \frac{1}{a} = \frac{-2}{a}$

$-\frac{\sqrt{3}\sqrt{a^2-4}}{2} + 1 = -2$

$\sqrt{a^2-4} = \frac{-6}{-\sqrt{3}}$

$a^2 - 4 = \frac{36}{3}$

$a^2 - 4 = 12$

$a^2 = 16$

$a = 4$ ($a > 0$)

From (1)

$$\sin \alpha = \frac{2}{9}$$

$$\sin \alpha = \frac{2}{9}$$

$$\sin \alpha = \frac{1}{2}$$

$$\alpha = \pi/6$$

max speed at centre of motion.

$$\ddot{x} = 20 \cos(5t - \pi/6)$$

$$-1 \leq \cos(5t - \pi/6) \leq 1$$

$$-20 \leq 20 \cos(5t - \pi/6) \leq 20$$

∴ max speed is 20 ms^{-1} and happens at $x=0$.

Changes direction when $\ddot{x}=0$

$$5t - \pi/6 = \pi/2, 3\pi/2, 5\pi/2, 7\pi/2, \dots$$

$$5t = \pi/6, 10\pi/6, 16\pi/6, 22\pi/6$$

$$t = \frac{2\pi}{15}, \frac{5\pi}{15}, \frac{8\pi}{15}, \frac{11\pi}{15}$$

$$\approx 0.4, 1.04, 1.67, 2.3$$

∴ Changes direction three times in first two seconds

$$(b) (i) (2+3x)^7 = \sum_{k=0}^7 {}^7C_k (2)^{7-k} (3x)^k$$

$$\therefore t_k = {}^7C_k 2^{7-k} 3^k$$

$$(ii) t_{k+1} = {}^7C_{k+1} 2^{6-k} 3^{k+1}$$

$$\frac{t_{k+1}}{t_k} = \frac{7! \times 2^{6-k} \cdot 3^{k+1}}{(6-k)! (k+1)!} \times \frac{(7-k)! k!}{7! 2^{7-k} 3^k}$$

$$= \frac{3 \times (7-k)}{2 \times (k+1)}$$

$$= \frac{21-3k}{2k+2}$$

$$(ii) \frac{21-3k}{2k+2} > 1$$

$$21-3k > 2k+2 \quad (2k+2 > 0)$$

$$5k \leq 19$$

$$k \leq 3.8$$

∴ $k=3$ gives largest coefficient

$$t_{3+1} = t_4 = {}^7C_4 \cdot 2^3 \cdot 3^4$$

$$= 22680$$

QUESTION 14

i) Let $\angle QAB = x^\circ$

$\angle CAD = x^\circ$ (L's in same segment)

$\angle AYA + 90^\circ + x^\circ = 180^\circ$ (L sum ΔAYO)

$\angle AXP = 90^\circ - x^\circ$

$\angle AXP + \angle AXP + \angle BXP = 180$ (straight line PAX)

$90 - x + 90 + \angle BXP = 180$

$\angle BXP = x$

$\therefore \angle BXP = \angle QBX$ (both equal x).

ii) $\angle DAX + \angle AXP + \angle PDA = 180$ (L sum ΔAXP)

$x + 90 + \angle PDA = 180$

$\angle PDA = 90 - x$

$\angle PDA = \angle PCB = 90 - x$ (L's in same segment)

$\angle AXP = \angle CXQ = 90 - x$ (vertically opposite L's)

$\therefore \angle CXQ = \angle PCB$ (both equal $90 - x$)

$\therefore \Delta QCX$ is isosceles

ΔBCX is isosceles

$\therefore BQ = XQ$ (L's opposite equal sides)

$XQ = CQ$ (" " " ")

$\therefore BQ = CQ$ (both equal XQ)

$\therefore PQ$ bisects BC .

b) i) when $y = -2$

$-2 = -30t \sin 5 - 5t^2$

$5t^2 + 30t \sin 5 - 2 = 0$

$t = \frac{-30 \sin 5 \pm \sqrt{900 \sin^2 5 + 40}}{10}$

$= 0.423$

$= 0.4$

ii) $\dot{x} = 30 \cos 5$

$\dot{y} = -30 \sin 5 - 10t$

$\frac{dy}{dx} = \frac{\dot{y}}{\dot{x}} = \frac{dt}{dx}$

$\tan \theta = \frac{-30 \sin 5 - 10t}{30 \cos 5}$

$= -0.2213$

$\theta = -12.48^\circ$

$\therefore \theta = 12.48^\circ$ or 167.52°

iii) $x = 30t \cos 5$ — ①

$y = -30t \sin 5 - 5t^2$ — ②

from ① $\Rightarrow t = \frac{x}{30 \cos 5}$ — ③

subst ③ into ②

$y = \frac{-30x \sin 5}{30 \cos 5} - \frac{5x^2}{900 \cos^2 5}$

$= \frac{-x^2}{180 \cos^2 5} - x \tan 5$

of the form $y = ax^2 + bx + c$

$\therefore$ parabolic

c) $\tan 26 = \frac{h}{XA}$

$XA = \frac{h}{\tan 26}$

$\tan 30 = \frac{h}{XC}$

$XC = \frac{h}{\tan 30}$

$\tan 28 = \frac{h}{XB}$

$XB = \frac{h}{\tan 28}$

i) $\tan \angle XAC = \frac{XC}{XA}$

$= \frac{h}{\tan 30} \times \frac{\tan 26}{h}$

$= 0.84478$

$\angle XAC = 40.2^\circ$

ii) $\frac{\sin \angle ABX}{XA} = \frac{\sin 40.2}{XB}$

$\sin \angle ABX = \frac{\sin 40.2 \times \tan 28}{\tan 26}$

$= 0.7037$

$\therefore \angle ABX = 45^\circ, 135^\circ$

$\therefore \angle ABX = 135^\circ$ or $\angle ABX > 49.8^\circ$

iii) $180 - 135 - 40 = 05^\circ T$ or $N5^\circ E$